

FROM MODEL DIFFS TO CAUSAL CONTROL

A shared sparse representation for explaining and testing behavioral differences between code models

BEHAVIOR



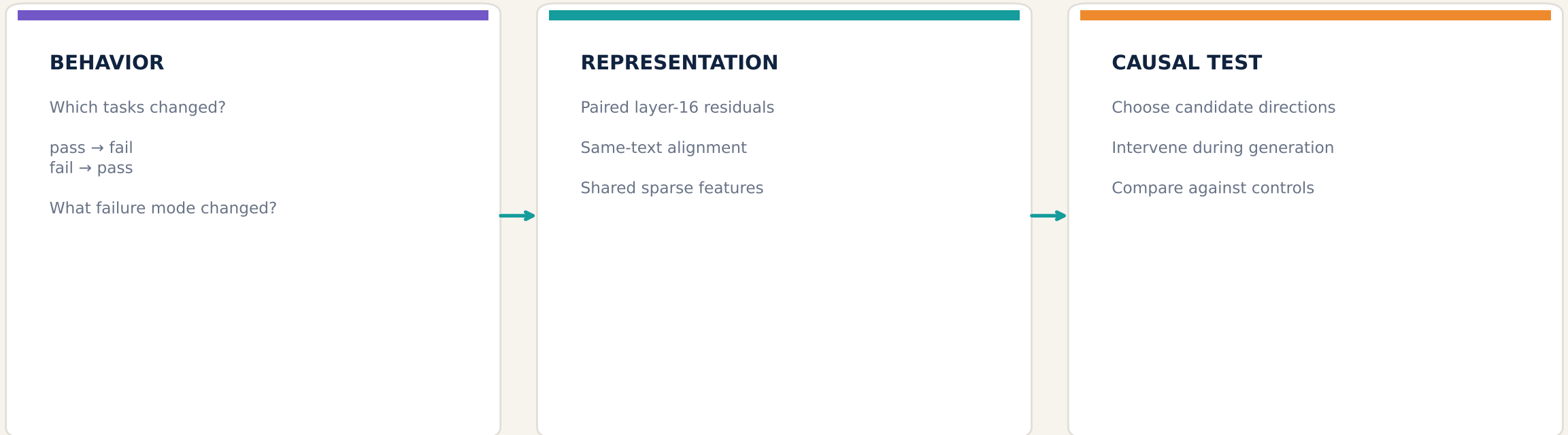
FEATURES



STEERING

RESEARCH QUESTION

Evaluation tells us what changed. CrossCoder asks which internal differences can explain—and control—it.



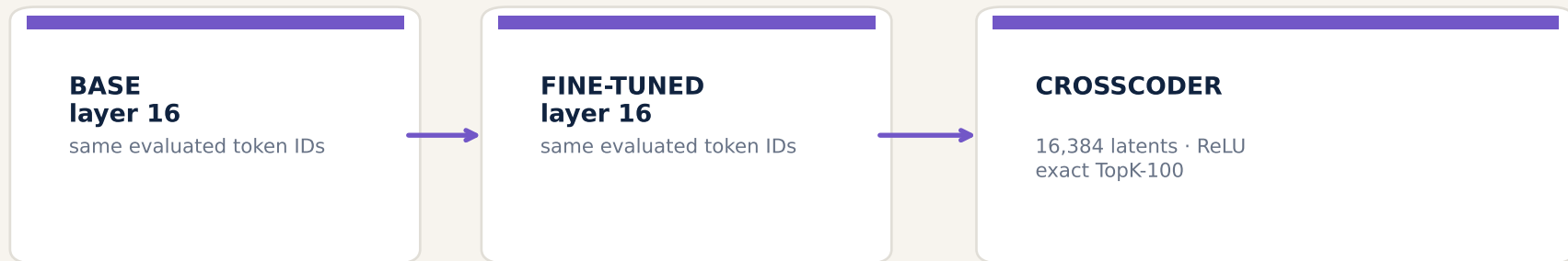
Behavior localizes the model difference. Sparse features turn it into a testable hypothesis.

ONE ANALYSIS FRAMEWORK, TWO VERY DIFFERENT MODEL CHANGE

The same same-text CrossCoder design spans fine-tuning and model merging

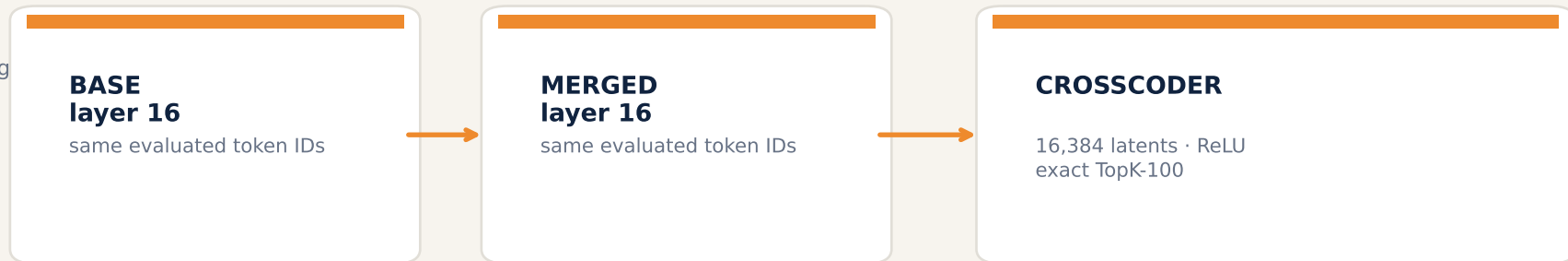
DEEPSEEK

specialization through fine-tuning



CODELLAMA

specialization through model merging



Different specialization mechanisms; identical analytical question: which shared latent differences track—and control—behavior?

THE TWO SPECIALIZATIONS MOVE PERFORMANCE IN OPPOSITE DIRECTIONS

Canonical extraction-v4 labels · same raw generations, audited reprocessing

DEEPSEEK · BIGCODEBENCH

268/1140

BASE

404/1140

FINE-TUNED

+11.93 pp

BigCodeBench
pass-rate change

CODELLAMA · BIGCODEBENCH

314/1140

BASE

27/1140

MERGED

-25.18 pp

BigCodeBench
pass-rate change

DeepSeek: 215 improvements · 79 regressions

CodeLlama: 4 improvements · 291 regressions

RECATCHER CONTEXT

Directionally consistent motivation: specialization can improve aggregate capability while still creating localized regressions.

Not a direct numerical replication: ReCatcher used 10 generations/prompt and different regression definitions.

WE FOCUS ON TWO ONE-SIDED BEHAVIORAL TRANSITIONS

Taxonomy turns aggregate model differences into semantically targeted cohorts

DEEPSEEK · 215 IMPROVEMENTS

base fail → fine-tuned pass

119/215: generated test/import contamination

48: wrong output or logic

16: missing name/import

32: other primary categories

Mechanistic focus: 80 contamination cases

CODELLAMA · 291 REGRESSIONS

base pass → merged fail

120/291: API/type mismatch

50: wrong logic/other runtime

43: edge case/exception

78: generation, import, syntax, commentary

Mechanistic focus: 50 wrong-logic/runtime cases

HOW WE CLASSIFIED

Rule/AI-assisted taxonomy + human spot checks

IMPORTANT QUALIFICATION

Primary = most salient observed error—not a unique root cause. CodeLlama failures are heterogeneous and oft

DEEPSEEK /5 · CONTAMINATION

BASE · FAIL

Correct-looking body continues into:

```
`#tests/test_task_func.py`  
`import pytest`
```

FINE-TUNED · PASS

Completes the function and stops after a short example.

First textual divergence $\approx 76\%$ of shorter code;
target boundary is late/post-solution.

CODELLAMA /490 · WRONG LOGIC/API CONTRACT

BASE · PASS

parses XML $\rightarrow$ writes JSON file $\rightarrow$ returns result

MERGED · FAIL

`return xmltodict.parse(s)`
omits the required file write side effect

First divergence $\approx 81\%$ of base code;
target region is the late function body.

FEATURE SCREENING

We rank stable behavioral contrasts—not raw activation alone



Minimum support

max(3 tasks, 10% of positives)

Interpretation

E/V is permutation signal-to-noise—not a z-score or p-value.

Positive and negative orientations are retained: either enrichment direction may support behavior-aligned steering.

EIGHT COMPLEMENTARY SCREENING CELLS PER MODEL

Transition type × comparison design × temporal scope

DEEPSEEK

REGRESSION

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

IMPROVEMENT

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

CODELLAMA

REGRESSION

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

IMPROVEMENT

ONLY 4 POSITIVES · EXCLUDED

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

Local = ±10% around the first normalized divergence—not a semantic error category.

WHAT THE SCREENING ACTUALLY RETURNS

Representative valid cells; E/V is the primary ranking signal

DEEPSEEK · IMPROVEMENT ASSOCIATION

Rank	Feature	Effect	E/V
1	3602	-0.021	-8.23
2	10168	-0.018	-7.67
3	5422	-0.005	-7.66
4	1982	-0.015	-7.53
5	1862	-0.008	-7.45
6	1650	-0.026	-7.28
7	5039	-0.088	-7.26
8	9580	-0.013	-7.24
9	5837	-0.010	-7.18
10	9537	-0.038	-7.15

CODELLAMA · REGRESSION ASSOCIATION

Rank	Feature	Effect	E/V
1	13147	+0.126	+5.24
2	8992	-0.011	-4.93
3	1840	-0.046	-4.77
4	16161	+0.127	+4.76
5	15471	+0.005	+4.75
6	10570	+0.188	+4.60
7	8138	-4.973	-4.51
8	7353	+0.228	+4.48
9	14359	-0.166	-4.46
10	2825	+0.338	+4.38

Examples—not a single global ranking. Candidate pools are formed by union and deduplication across valid cells.

EIGHT PRE-DECLARED TOP-10 LISTS DEFINE THE $\alpha=3$ CANDIDATE POOL

Association vs paired model $\Delta \times$ global vs error-local scope

DEEPSEEK · IMPROVEMENT

ASSOC · GLOBAL	ASSOC · LOCAL	PAIRED · GLOBAL	PAIRED · LOCAL
3602	900	11172	9009
10168	14653	1767	5456
5422	2021	16044	309
1982	7279	10532	12591
1862	16331	9632	9982
1650	2582	7027	6927
5039	12591	1395	1126
9580	2380	6565	900
5837	10257	14818	13220
9537	15715	6911	6356

40 nominations → 38 unique features

2 feature(s) appear in more than one list

CODELLAMA · REGRESSION

ASSOC · GLOBAL	ASSOC · LOCAL	PAIRED · GLOBAL	PAIRED · LOCAL
13147	4837	8138	3612
8992	2861	3612	8309
1840	12780	12253	5642
16161	4313	1840	6214
15471	10879	11227	13428
10570	10111	14765	1231
8138	8558	4148	13934
7353	71	2286	8965
14359	9579	12573	11222
2825	8545	12643	354

40 nominations → 37 unique features

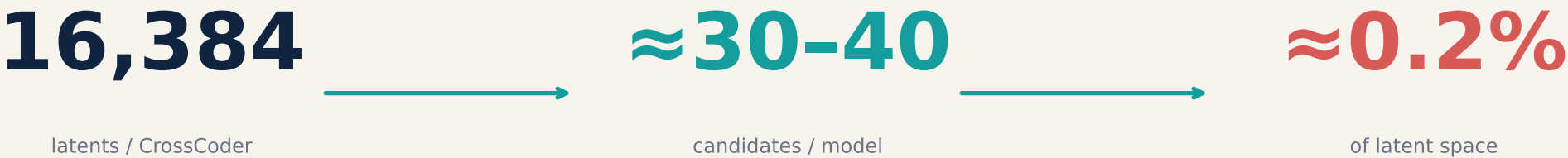
3 feature(s) appear in more than one list

NEW $\alpha=3$ SWEEP: exactly 38 DeepSeek + 37 CodeLlama features · no post-hoc additions

Hard gate: baseline must reproduce the original raw completion byte for byte before any steered arm runs.

SCREEN FIRST; TEST CAUSALLY AT A COMMON OPERATING POINT

The statistical shortlist reduces the intervention search by ~99.8%



WHY $|\alpha| = 3$?

Earlier dose-response work suggested a practical exploratory point: visible causal changes below the most disruptive high-dose regions

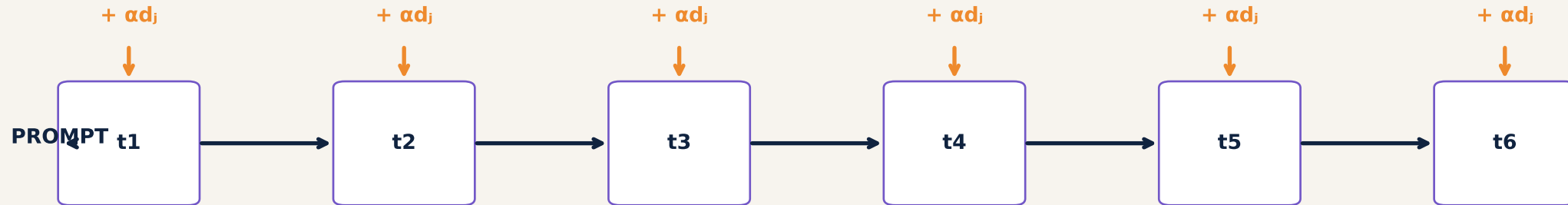
WHY STANDARDIZE?

One disruptive high-dose region of candidates comparable before full dose curves, reverse-model t

$|\alpha|=3$ is an operating point—not an independently validated optimum.

OUR CURRENT INTERVENTION IS CONTINUOUS STEERING

The selected decoder direction is added at every autoregressive generation step



CURRENT PROTOCOL

Layer 16 · last residual position
feature decoder on intervened model side
applied through the full generation

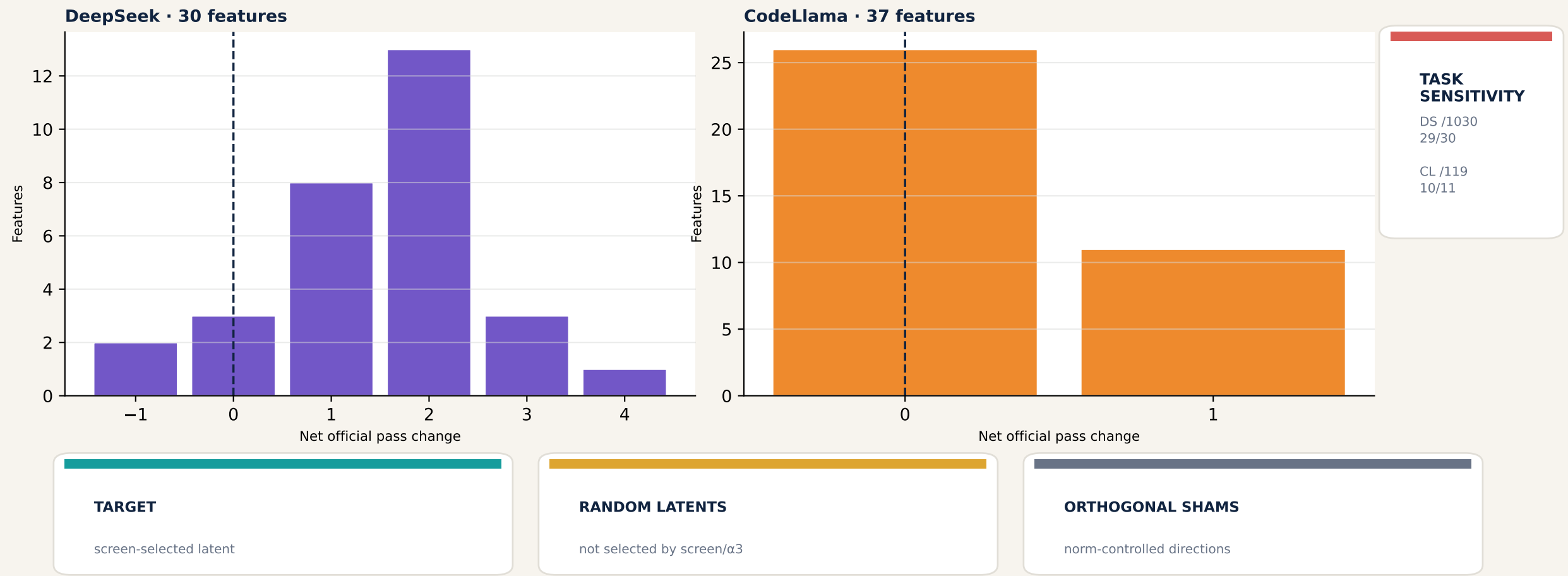
WHAT IT IS NOT

Not a one-token intervention
not activation-gated or clamped
not conditional on natural activation

This tests continuous control of the generation trajectory—not whether one naturally active token alone caused the error.

THE $\alpha=3$ SWEEP REVEALS BOTH MODEL AND TASK SENSITIVITY

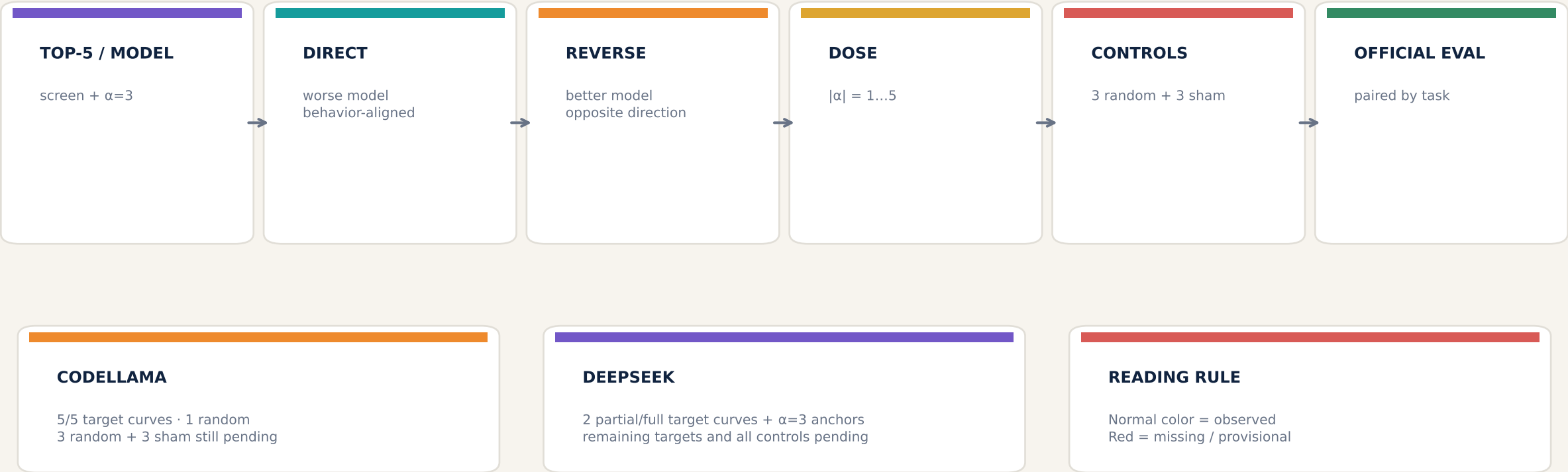
Official pass changes motivate—but do not replace—random and sham controls



Controls are always good practice; concentrated successes make their necessity especially visible here.

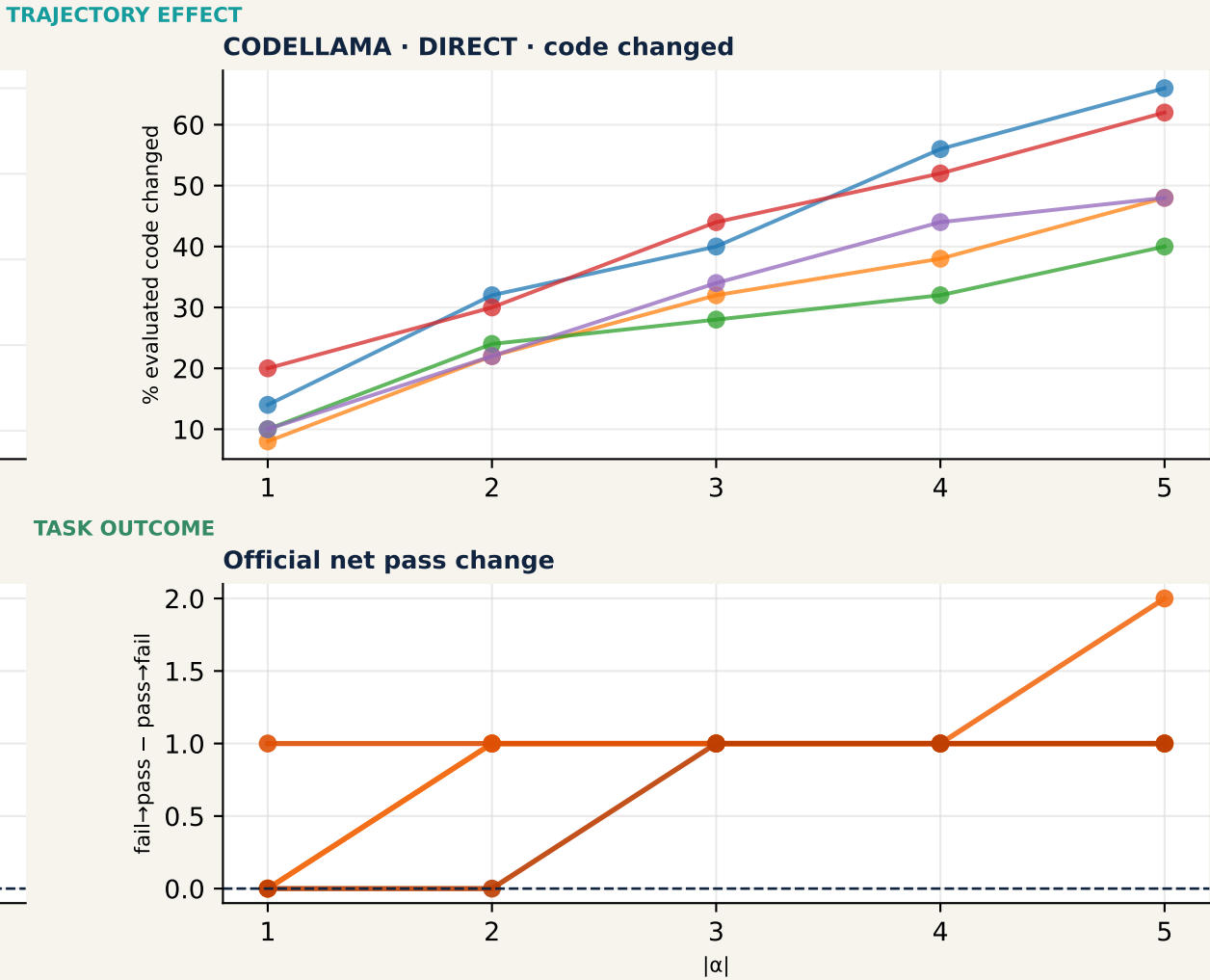
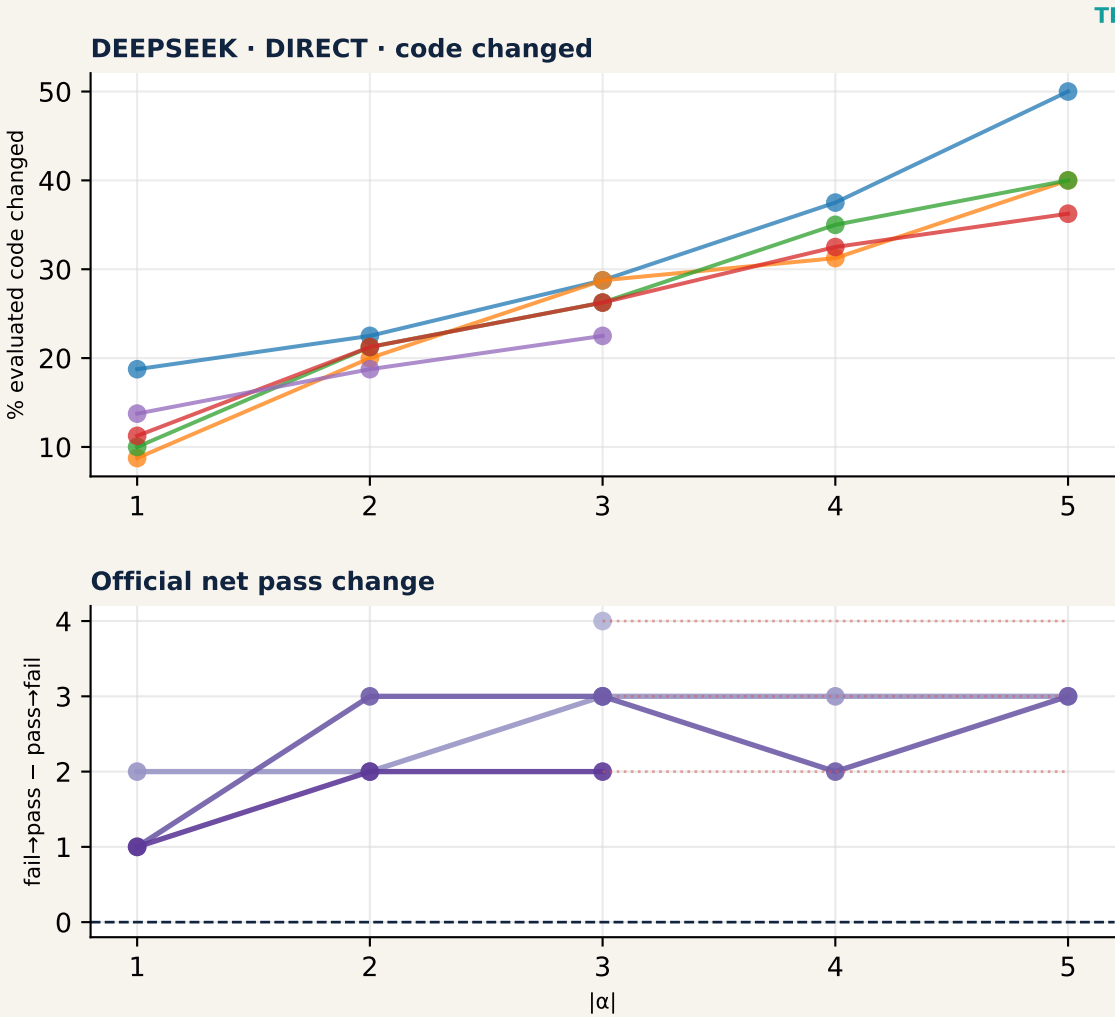
BBASV TURNS ASSOCIATIONS INTO CONTROLLED CAUSAL TESTS

Bidirectional Behavior-Aligned Steering Validation



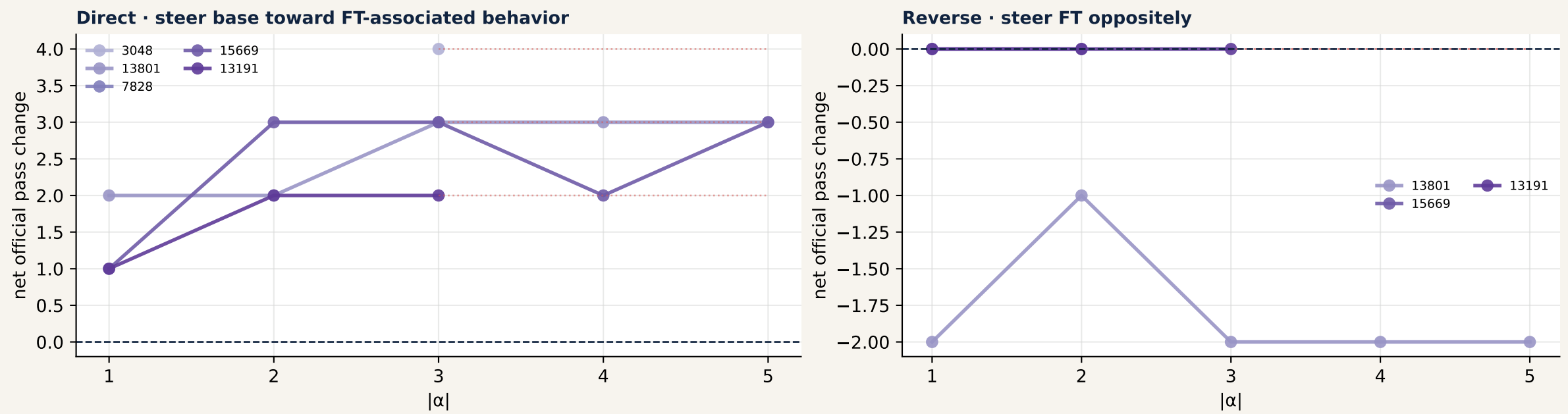
STEERING CHANGES TRAJECTORIES MORE OFTEN THAN OUTCOMES

Two aligned endpoints: evaluated-code change and official fail↔pass



DEEPSEEK: EARLY RESULTS SHOW POSITIVE BUT TASK-CONCENTRATED

Direct baseline 15/80 · reverse baseline 57/80 · 25/45 non-baseline target arms evaluated



FAVORS THE STORY

13801: +2,+2,+3,+3,+3 direct
and -1/-2 net in reverse

QUALIFIES IT

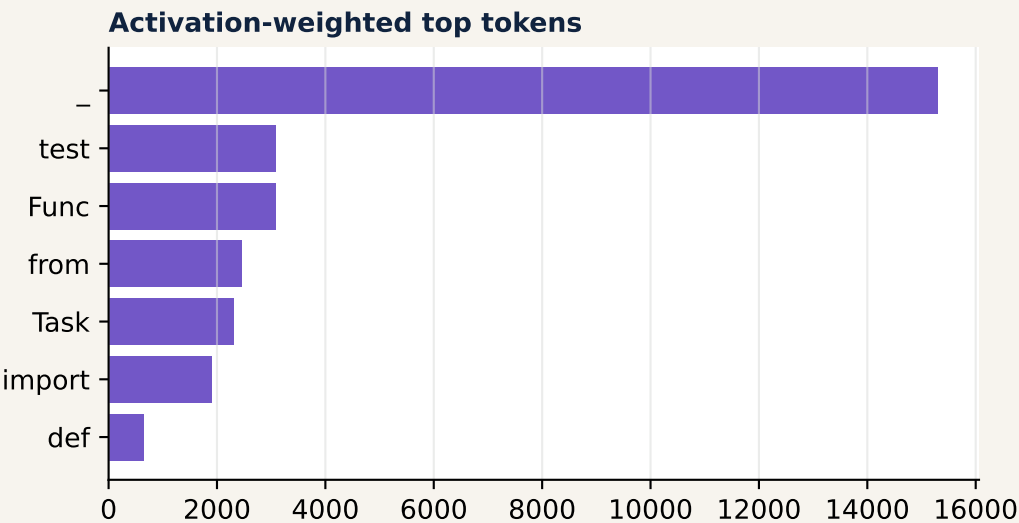
/1030 and /706 recur across targets:
part of the gain is task susceptibility

PENDING

3048 and 7828 dose curves
all random and sham controls

DEEPSEEK FEATURE 3048: A COHERENT TEST/IMPORT-BOUNDARY CAND

Semantic evidence and one clean $\alpha=-3$ fail→pass example



BIGCODEBENCH/762 · FAIL → PASS

BASE: broad try/except + appended pytest suite

STEERED $\alpha=-3$: direct implementation + return zip name;
post-solution tests removed

Official transition: fail → pass

MEANING

test/import/function-boundary contexts

FAILURE MODE

post-solution contamination and
over-defensive implementation

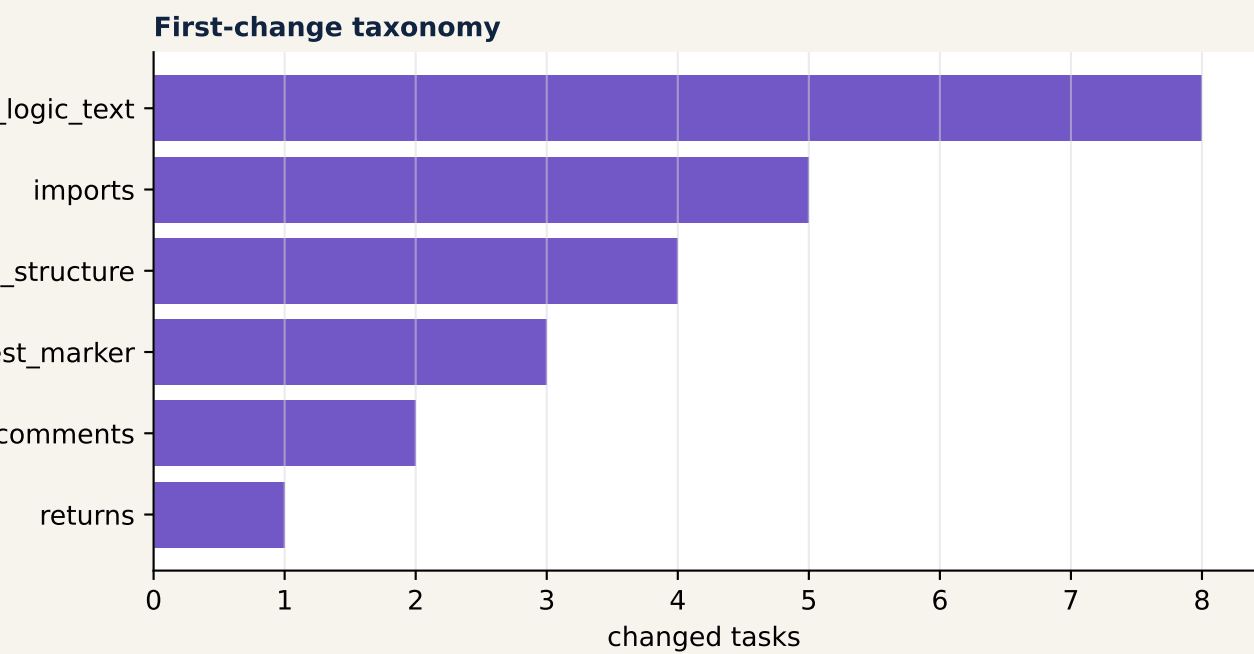
CAUSAL CHANGE

removes tests and simplifies body
in a task that becomes correct

Interpretation is coherent, but 3048 specificity versus controls remains PENDING.

FEATURE 3048 OFTEN CHANGES THE RIGHT REGION WITHOUT SOLVING

At $\alpha=-3$, 23/80 evaluated codes changed; only 4 became official passes



WHAT CHANGED

imports/tests/comments/function structure
frequently move—the expected semantic neighborhood

WHY MOST STILL FAIL

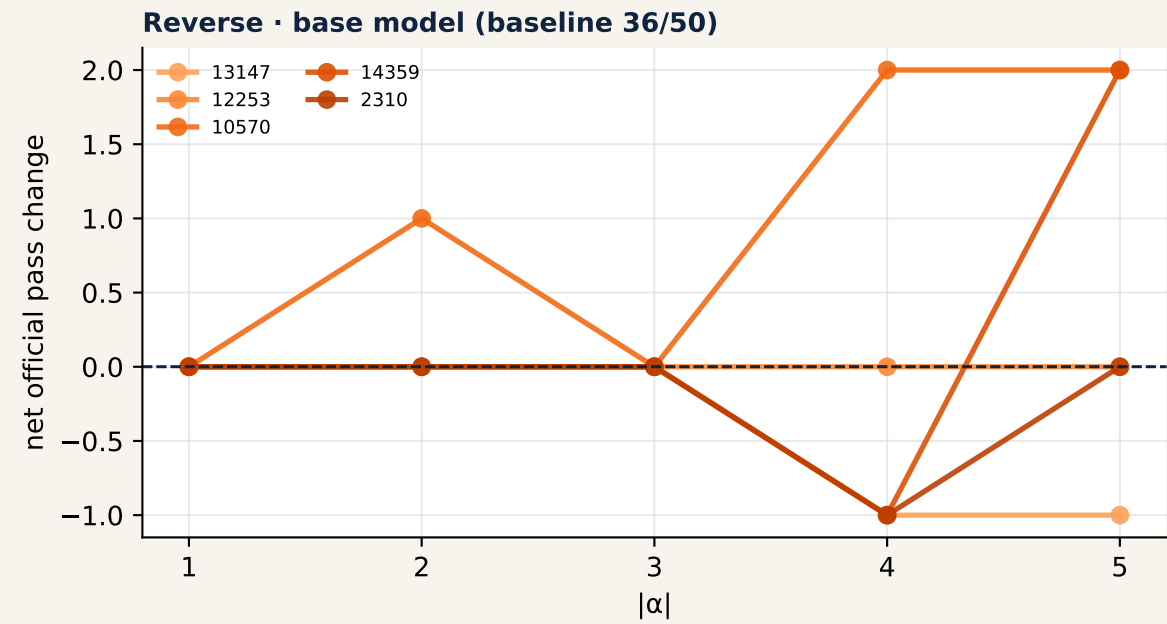
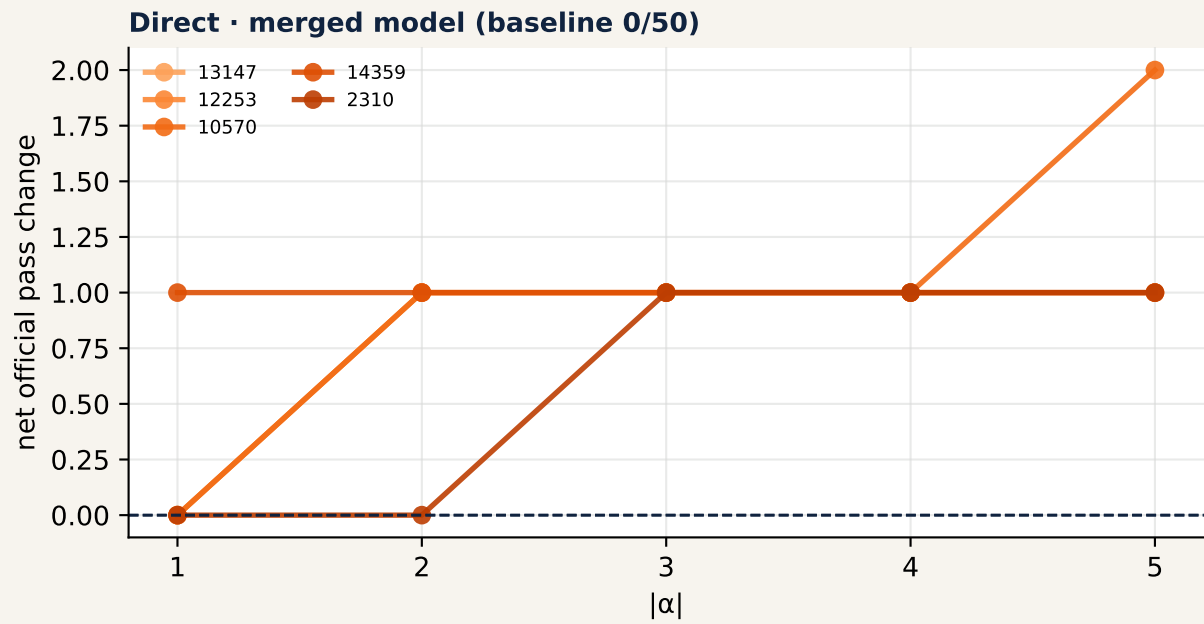
Removing contamination cannot repair every
underlying logic/API/runtime defect

23 code changes → 4 fail→pass

Causal scope ≠ complete task solution

CODELLAMA: COMPLETE TARGET CURVES SHOW WEAK OUTCOME SPEC

All 50 target direct/reverse arms evaluated · one random latent evaluated · shams pending



TARGET RESULT

mostly 0 or +1; 10570 reaches +2
at $|\alpha|=5$

CONTROL RESULT

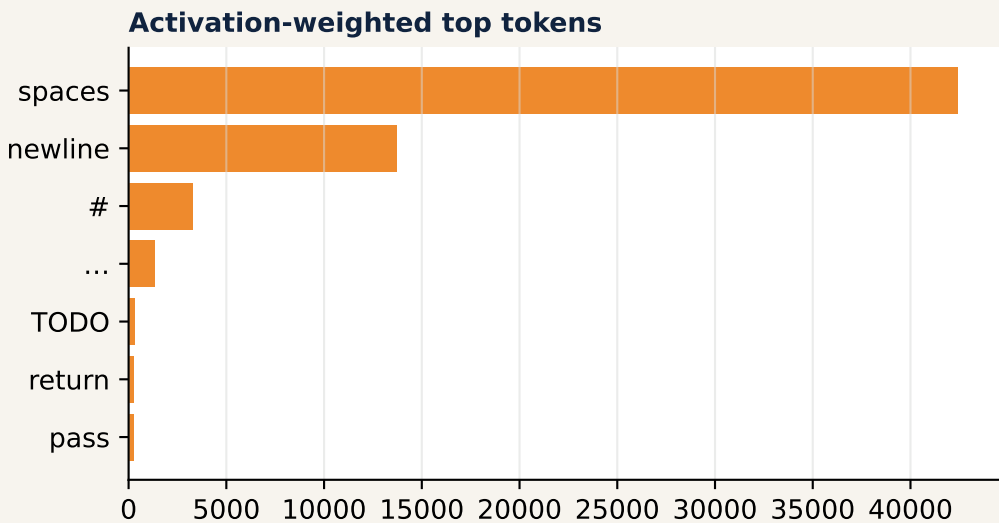
random 13879 reaches +1 direct
and +2 reverse

CURRENT READING

trajectory control exists; outcome
specificity is not established

CODELLAMA FEATURE 13147: A CLEAN RETURN-VALUE REPAIR

Task /490 is corrected from $|\alpha|=2$ through 5 and differs from the common /119 response



BIGCODEBENCH/490 · FAIL → PASS

BASE: parses XML and writes JSON—but returns nothing

STEERED: adds `return result` to both generated functions

Stable at $\alpha=-2,-3,-4,-5$

MEANING

incomplete function / placeholder /
return-boundary contexts

FAILURE MODE

required return value omitted

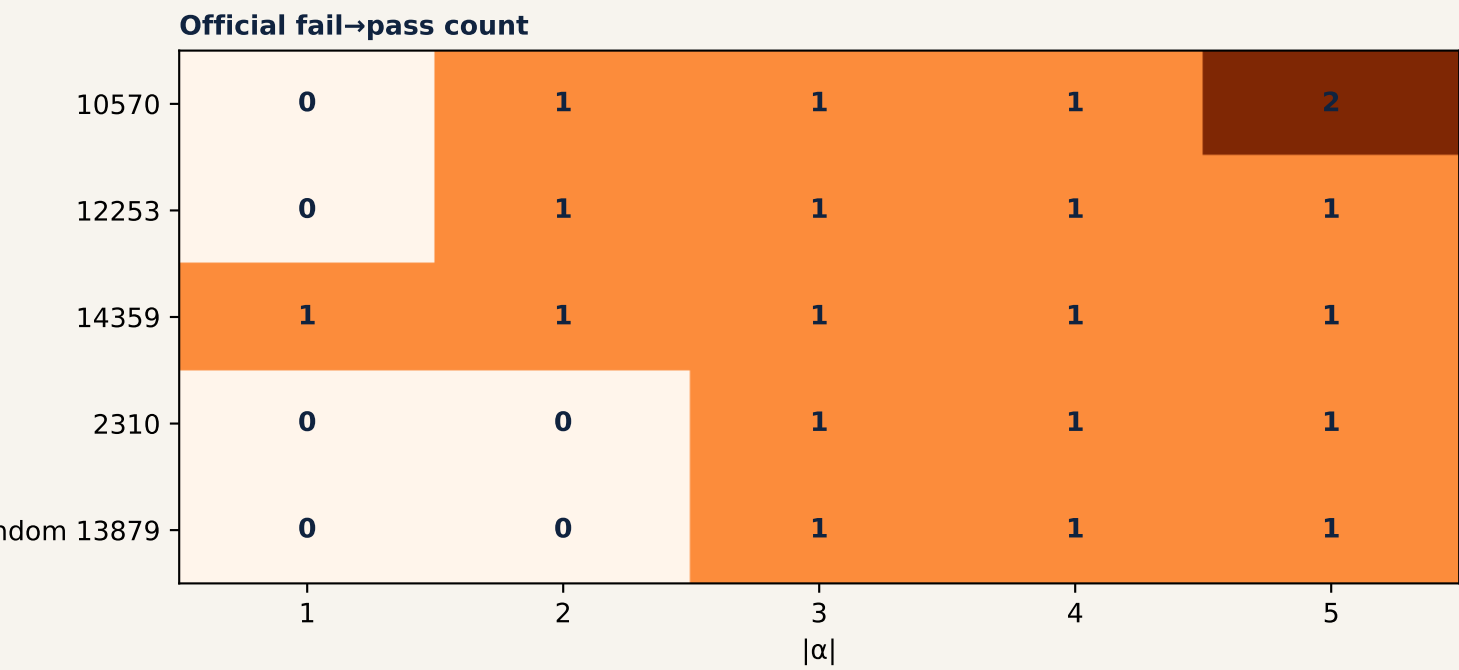
REVERSE TEST

base loses 1 pass only at $|\alpha|=4-5$

Promising mechanistic example; matched random/sham comparison for /490 remains PENDING.

CODELLAMA ALSO EXPOSES A NON-SPECIFICALLY STEERABLE TASK

/119 is corrected by four targets—and by the random feature 13879



AGAINST SPECIFICITY

A random latent reproduces the same /119 transition at $|\alpha|=3-5$

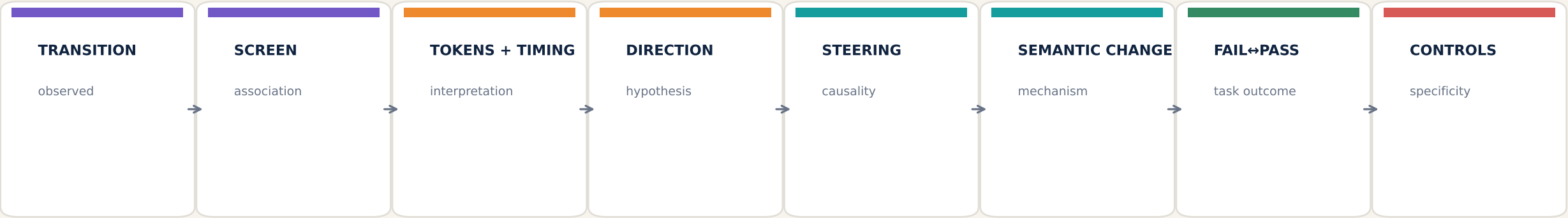
FOR THE METHOD

The control catches exactly the overclaim BBASV was designed to detect

Causal change? YES Feature-specific mechanism? NOT YET

THE EVIDENCE CHAIN MAKES EACH CLAIM AUDITABLE

Association, interpretation, causality, and specificity are different evidential layers



DeepSeek 3048 ✓ ✓ ✓ ✓ ✓ ✓ ✓

controls pending

CodeLlama 13147 ✓ ✓ ✓ ✓ ✓ ✓ ✓

partial controls

The methodology does not treat every fail→pass as feature-specific; it tests where the evidence chain breaks.

WHAT THE PARTIAL RESULTS SUPPORT—AND WHAT THEY DO NOT

Current claims are deliberately tiered

SUPPORTED NOW

- screening compresses 16,384 latents to testable candidates
- selected features causally alter generation trajectories
- some interventions create official fail→pass transitions
- 3048 and 13147 have semantically coherent examples
- controls identify task-level susceptibility

NOT YET SUPPORTED

- all model regressions/improvements are feature-caused
- every successful target is feature-specific
- CodeLlama targets outperform generic perturbations
- generalization beyond selected cohorts
- robustness across seeds

Best current causal claim

Selected CrossCoder features can exert directional, semantically coherent control in selected tasks.

Specificity is a result to be demonstrated—not assumed from screening or a single correction.

A REUSABLE METHODOLOGY FOR CAUSAL MODEL DIFFING

The output is a falsifiable mechanism hypothesis—not merely a feature ranking

1 Paired behavioral transitions

2 Same-text sparse representation

3 Conditioned multi-cell screening

4 Meaning + timing + failure mode

5 Directional causal hypothesis

6 Dose-response steering

7 Official paired evaluation

8 Random + matched sham controls

9 New seeds / held-out validation

Each stage can reject or refine the hypothesis before a broad mechanistic claim is made.

FROM MODEL DIFFERENCES TO TESTABLE CAUSAL MECHANISMS

The partial BBASV results already show both the promise and the discipline of the framework

Model differences are not only observable in behavior.

They can be localized in a shared latent representation,

interpreted in context, and—in selected cases—

causally manipulated.

THE CONTRIBUTION

A testable path from behavioral contrast → sparse feature → semantic hypothesis → controlled intervention → specificity test

PENDING: complete DeepSeek controls, CodeLlama sham/random controls, and multi-seed validation.